

Block

2

RESEARCH METHODS- QUANTITATIVE

UNIT 6

Data Collection Sources

UNIT 7

Survey Methods

UNIT 8

Content Analysis

UNIT 9

Experimental Method



UNIT 6 DATA COLLECTION SOURCES

Structure

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6.0 INTRODUCTION

Research in communications is a complex process. That is why this course is organised to assist you to proceed, in a systematic way, to undertake research in the subject. The first block of the course was about introducing you to research and about helping you to decide what kind of research you wish to take, how to define your problem, how to draw a sample, and what is meant by a review of literature. From Block 2 onwards, we shall take you through the entire process of research and explain how to find information, design tools based on quantitative and qualitative approaches, and to analyse data thus collected.

Media research is assuming increasing importance as information is an important part of our daily lives. All organisations, especially those in the media, need data. Data, which are then converted into information and knowledge, goes toward enhancing a given topic and helps in decision making. Data can consist of background information about a celebrity; data can also consist of voting patterns in a previous election. However, not everything published, online or offline can be relied upon to be called 'data'. In understanding data for research, it is important to distinguish data by the source from which it is derived and by the way in which it is gathered.

The most essential component of any research, including mass communications, is that it must be conducted according to sound principles; else, it will not yield accurate results and will be a mere wastage of time, efforts and resources. Such research conducted must use effective data collection sources to ensure that the research study is based on accurate and valid information.

The term 'data' as you may be aware is plural. 'Datum' is the term used for a single piece of data. Our focus is on data. **Data collection** is the process where the information is gathered and measured on variables of interest. It is important at all stages of a research process. Developing research questions and hypotheses and their subsequent tests need good data.

6.1 LEARNING OUTCOMES

After completing this unit, you should be able to:

- describe various primary sources of data collection;
- discuss the secondary sources of data collection;
- explain how to access and use various online databases;
- trace the connection between different forms of research and data collection; and
- store and save your data.

6.2 PRIMARY AND SECONDARY DATA

Research, as you are aware is based on data, not on opinions and hunches. Therefore it is extremely important that the data collection process is accurate and appropriate. Sometimes, even the approach to a research problem can depend on the preliminary information that is collected. What data we collect and how we collect are critical so that we can undertake the following activities:

- frame the research question in a suitable way
- collect relevant data
- avoid having data gaps
- save valuable time and resources
- make accurate, reliable, and sound analyses
- help future researchers navigate the data available
- guide future researchers in further investigating a topic.

As discussed in the previous units the Sources of Data can be of two types - Primary and Secondary. Let us understand them in greater detail.

Data are data which are collected fresh and or the first time. For example, although data may already exist in government databases, one may still want to go to a village and collect fresh information on the number of disabled persons and the types of disability they have. What is then collected is primary data, original in character. During a census every ten years, the government collects primary data; and every few years, government conducts the National Family Health Surveys. Such data collected from the field are primary and they are analysed before being presented to us in reports. When we read the reports, we are looking at secondary data and not at primary data.

There are several methods of collecting primary data. We may collect the data through surveys, through observations, through discussions and interviews with experts and with respondents in our selected research area. We may also collect primary data through photographs, audio, and video recordings that we take when in the field. The key characteristic of primary data is that it is originally collected.

Therefore, primary data are:

- Collected at the time when an event has occurred.
- Data are in original form and are not published anywhere else and have not been interpreted or translated.

- Data thus collected help to create original documents because the origin is the research conducted in the field.

Secondary data on the other hand, are data which are already available and have been collected and analysed by someone else. Such data may have already gone through statistical procedures and analysed for example, Census reports, and national readership surveys. In these cases, an agency has collected data from the field, tabulated and analysed it and has presented it in the form of tables and charts, and reports for others to use. For example, we may refer to the findings of a national readership survey by saying “According to data reported in the latest National Readership Survey, 2018” while reporting findings. Secondary data are also available and collected from a number of sources such as publications of central and state governments, annual reports, technical and trade journals, books, research reports. Some such data may have been published in journals, others may be unpublished reports.

The key factors to remember about secondary data are:

- Secondary data are not original.
- Secondary data have already gone through some analysis, so the use of secondary data means that we are examining already examined data and conclusions.
- Secondary data may have been taken from an original source, but have been repackaged for a set of objectives which may be different from our research.
- Secondary data may have been analysed differently.
- Use of secondary data is a cost effective method of data collection but such data must be reliable, valid and accurate. For example, Wikipedia is not considered as a reliable source of secondary data for academic research.

There are some advantages of using secondary data in research. The use of secondary data is cheaper, time saving, and is a quick way of gaining an understanding of the subject at hand. Sometimes, reanalysis of secondary data may yield different results from the original.

But there are also limitations to secondary data. Because such data come from already published reports, information may be outdated. For instance, much of the data in the Census Report 2011 is dated because the report gives information about the situation in 2011/2012, but cannot be counted upon to be accurate in 2018. This is also the case with many field based reports, because there is always a time lag between the collection of primary data, the analysis and the publication as a report in respected journals. Sometimes this time lag can even be three or four years long.

Requirements of Secondary Data

How does one then make the necessary choices about when, where, and how to use different data sources? We suggest some simple ways of deciding when to use secondary data, when to use primary data, and what are the yardsticks for evaluating secondary data.

Most research in mass communication uses a combination of both primary and secondary data. The extent to which a data source is used depends on the kind of research being undertaken. Then, the first issue that comes to mind is: what is

the kind of research we are undertaking. Let us examine this in terms of the classifications of research that were presented in Unit 2.

There are four requirements in selecting secondary sources of data.

- 1) **Availability:** Secondary data must be available. For instance to take the example of the physically challenged persons cited earlier in this unit, if details about the physically challenged in a village are not available in the census or other documents, then we would have no choice except to collect it afresh. We may also get, at most, a number of physically disabled persons, but may not know specifically who they are in a family.
- 2) **Relevance:** Data must be relevant to the problem under study. Do not use data sources which are not relevant.
- 3) **Accuracy:** Secondary data must be accurate and reliable. There must be clarity on how the data were collected, what methodology were used; what the margin of error was and whether the sources were dependable.
- 4) **Sufficiency:** There must be enough data available. If enough data are not available, primary data collection is a must.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) Define Primary and Secondary Data.

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- 2) What are the important factors in choosing secondary data?

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6.3 SOURCES OF SECONDARY DATA

Let us look in more detail about the sources of secondary and primary data especially in terms of the processes that are followed in carrying out research studies. Communication is an interdisciplinary subject therefore we often research topics which are not just about media; but also about many other disciplines. This is why we must cast our nets wide in our search for secondary data and must look at a range of sources for information.

When we are about to embark on a research project or study, many things are unclear. At this stage, we are still looking at a broad area and we do not know

what research on the given topic is available, who has done research before, and where is the information to be found that will guide us to proceed. What are the secondary data sources that you are likely to search for? Some of these are:

- Library resources
- Previous research reports/ Project reports
- Historical accounts
- Books and materials
- Official/non-official records
- State and district gazetteers, Census documents, Statistical reports
- Autobiographies, letters, diaries
- Documentary films, Programme transcripts
- Photographs, Maps
- Newspapers, Magazines
- Discussions with experts /People

This is the first step and it is here that secondary data and secondary data sources are most important.

6.3.1 Internet Search

In today's world, however, an Internet search is the first place to start with. This is because most publications are now indexed online, and hard copies of the indexes may or may not be available in a library. Hard copies are published year wise, while the online index allows you to search across several years in one place. Use a search engine like Google and use key words to search for the topic under study. The search engine will throw up vast amounts of material, some usable, some not usable.

For academic research, some Internet based secondary sources in the field of social sciences include Google Scholar, the Social Science Citation Index, The Arts and Humanities Citation Index, The Communication Initiative Network, ERIC, JSTOR, Science Direct, Infilbnet among others.

Google Scholar

When you first search Google for a topic, the search engine will first show you what articles are listed and emerge in Google Scholar. Google Scholar is web search engine which specifically focuses on academic and scholarly articles. With a single search, you can search across many sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites. Google Scholar will then provide you with the link telling you where the book or article exists, sometimes provide you a link through Google books (where you can read the book or article) and also show you how to access it. It is a good place to start your hunt for secondary sources. You will also find out if the full article is available free of cost, on payment, who the copyright holder is, and how the article is to be cited in bibliographic citation format needed in academic publications.

Social Science Citation Index

The Social Science Citation Index (<http://mjl.clarivate.com/cgi-bin/jrnlst/jloptions.cgi?PC=SS>). covers about three thousand journals across 50 disciplines in the social sciences. You can search the index by title, author, subject, and journal or ISSN number. The ISSN number is the international classification of journals. You can view a list of the all the journals included in the Index and search for materials in both current and older journals. If you do not have any preliminary information, search by subject, using key words.

The Arts and Humanities Citation Index

The Arts and Humanities Citation Index (http://mjl.clarivate.com/scope/scope_ahci) is similar to the Social Science Index and is available from the same online provider. This index covers more than 1700 arts and humanities journals and has both indexing and abstracting. You can search by topic or title or journal.

The Communication Initiative Network

The Communication Initiative Network (<http://www.comminet.com/global/content/overview-communication-initiative>) is a network of more than 100000 individuals and organisations working in the field of communication and development. You can search their database for research papers, case studies, knowledge summaries on important topics. The initiative provides free access to a vast repository of materials on various aspects of communication and development.

ERIC (Education Resources Information Centre)

ERIC is an online digital library of education research and information sponsored by the U.S. Department of Education. (<https://eric.ed.gov/>) ERIC provides access to 1.5 million bibliographic records (citations, abstracts, and other pertinent data) of journal articles and other education-related materials, with hundreds of new records added every week. The ERIC Collection, begun in 1966, contains records for a variety of publication types, including journal articles, books, research syntheses, conference papers, technical reports, dissertations, policy papers, and other education-related materials ERIC also has a connection of grey literature, i.e. materials produced by individuals and organisations outside the normal means of distribution such as journals and published books.

JSTOR

JSTOR (<https://about.jstor.org/mission-history/>) is a digital library for scholars, researchers, and students which covers more than 2500 journals from 57 countries. JSTOR provides free and low cost access to academics and scholars in the development countries through partnerships with institutions and philanthropic institutions. Currently, JSTOR has in its digital collection more than 10 million academic journal articles, 50000 books, and 2 million primary source documents. In 75 disciplines. After Google, this is a good place to begin the search for secondary data.

Science Direct

Science Direct (<https://www.sciencedirect.com/>) has a huge collection of Social Sciences and humanities journals and books, highlighting historical context,

current developments, theories, applications, trends and more. When you go to Science Direct, you may think that there are no materials in mass communication. However, what you must understand is that because mass communication is an interdisciplinary subject, secondary sources of data in other subjects like Psychology, Education, etc. are also important. Also, if you are looking for research on theme areas in development, such as health communication, there will be a wealth of articles available through this digital repository of articles and journals. Currently, more than 250,000 articles in Science Direct, are open access, made free to read, download and use keeping in line with the author's licensing.

Inflibnet

Inflibnet, an Inter-University Centre of the University Grants Commission is India's digital network of libraries and resources. It was set up to provide support to academic research and learning through cooperative facilities among universities and other academic institutions. Inflibnet has many activities, one of which is E-ShodhSindhu (<https://www.inflibnet.ac.in/ess/about.php>). ShodhSindhu provides current and archival access to more than 15000 core and peer reviewed journals and a number of bibliographic, citation and factual databases in different disciplines from a large number of publishers and aggregators. Students and scholars can access these databases through member institutions including centrally funded institutions, universities and colleges. Through its e resources, you can access journal articles and reports and databases. Through its activity IndCat, Inflibnet also provides access to an online catalogue of holdings (books, journals, theses and dissertations), in Indian University Libraries. This online catalogue is very helpful for students and researchers as it informs you in which library a particular book may be available. An interlibrary loan of materials is facilitated through Inflibnet.

The sources for secondary data given above are largely free databases. However, on the Internet search, you may find many articles freely available in the public domain and those that are copyrighted and not accessible unless you make a payment. In developing countries like India, the cost of paying US\$35 just to view an article can be very prohibitive. This is where access to a library becomes important.

6.3.2 Library Search

Before the age of the Internet, the library was the place where a researcher would go to collect data. The library is a storehouse of books, journals, indexes, reports, and documents, newspapers, magazines and even unpublished documents. Not only do libraries hold published materials, many libraries also hold collections of old manuscripts, scrolls, both in hard documents and within their microfilm and microfiche and e resources sections. Libraries are also able to access e journals and books through interlibrary loan facilities.

Unless you have a borrowing facility as a library member, most libraries, you will find, will allow you to read the books and journals there in the library itself. The first place to start in the library is the card catalogue. Many libraries have both online searches as well as drawers full of library cards organised in an alphabetical manner. Search the card catalogue first by the topic of your study. This is because, at this stage, you may still be unaware of the names of scholars

or titles of books on the topic under study. You can also search by author and/or title of the book. Books are arranged on shelves by the order in which they are numbered in the library catalogue. Usually it is the Dewey Decimal Classification (DDC) system or the Library of Congress (LCC) system which a library follows.

After your book search, proceed to the Reference section of the library. This is where reference materials, such as Encyclopaedia, Yearbooks, and even the hard copies of the Citation Indexes mentioned in the earlier section are housed. Once you find the journal article reference, go to the journals section of the library. This is where journals are kept and you may find the issue of the journal there. Newspaper articles are usually kept in the microfilm or microfiche collection of the library, because it is difficult for a library to stock print copies. If you cannot find what you want, ask for help, and the librarian will tell you whether it is available in the library or at the NASSDOC library or can be obtained on an interlibrary loan from some other library.

You may also like to explore the NASSDOC library of The Indian Council of Social Science Research. NASSDOC provide documentation and information support to researchers in social sciences and contains a huge collection of documents for use. Scholars working in autonomous research organisations, universities, government agencies, can use this facility to obtain data very easily.

Newspaper and Magazines

In media research studies, newspapers and magazines can play an important role in collection of secondary data. With the development of internet, e-papers are also getting popular. Thus, getting information on any topic is very easy. There are opinions, columns, news stories, editorials, features, reports and essays that can be referred to for research. However, it is advisable to use newspapers and magazines carefully and only for some background information, and to get a time reference for an event. Newspapers and magazines are not recognised as valid sources for academic research. Equally, articles in some open access journals (which are not peer reviewed), Wikipedia sources, and blogs of individuals are not recognised as valid sources for academic research.

Much of secondary sources of data are consulted at the formative stage of a research project, in fact, much before the researcher goes to the field. Secondary sources of research form the heart of the Review of Literature discussed in Unit 5. In turn, the review of literature forms the basis for developing research questions and hypotheses, and which serves later to link the current research to a body of theory on the subject. Being able to identify, sift the useful from the irrelevant and find knowledge gaps which need to be filled, needs that we understand how to find secondary data and how to use it. And finally, because our sources of secondary data have to be reliable, we need to know how to identify reliable secondary data.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Name a few freely available Internet sources for secondary data.

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2) What is JSTOR?

3) Name two salient features of Infilbnet which are useful for Indian academics and researchers?

6.4 SOURCES OF PRIMARY DATA

Primary Data are original research data that are collected by an organisation or an individual i.e. first-hand data that are collected. It is often assumed that, since the data are collected by the researcher him/herself, such data are more reliable. This is not necessarily so, unless rigorous and systematic steps are taken while collecting data. How to collect accurate, unbiased, and reliable and relevant primary data is discussed at length in various units of this course. At this stage, we will simply discuss what are the various ways in which primary data can be collected. Collecting primary data is often time consuming, and also more costly to carry out.

Primary data are also referred to as **raw** data. However, once such raw data are cleaned and processed by one team of researchers, the data become secondary data for another group of researchers.

What are the characteristics of primary data?

- It is data collected in its original form.
- Such data have not been analysed or published anywhere else. It may be fresh data collected in the field, relevant only in terms of a specific date, e.g. exit survey data during elections.
- Such data are always specific to the researcher's needs.

What are the sources of primary data? There are many kinds of primary data. These include original letters, historical documents, photographs, audio and video recordings, archives, surveys and observations can also be original data. We will discuss different kinds of primary data in mass communication terms of the different objectives of a given research effort.

If the objective of your research is to understand the role of the press in the Independence struggle in India, you will definitely look for books and articles

written about the topic. This is **historical research**. You may also go to original pages of newspapers, and original audio recordings. For example, in a pioneering research on the role of Gandhi and Mao as communicators, Dr. Kusum J. Singh (1980) used the original recorded messages of Gandhi and Mao during the period 1942-1944. She listened to the recordings, transcribed them and analysed them.

For her research, these original recordings were primary data.

You may study the original discussions of the Constituent Assembly to see what was discussed when India's fundamental right to speech was included in the Constitution. You may also study parliamentary proceedings, to understand what was discussed in the Parliament. You may study the diaries, personal letters, photographs, and other communications made by leaders and historical figures. In all cases you would use original materials. They include letters, life histories, memoirs, diaries, biographies and autobiographies etc. These help in getting information from the person himself and thus are reliable enough to be considered as primary data. If you are doing legal research, you would study case files, legal decisions and previous case histories and decisions. These documents give information regarding happenings in the past and are written by an individual to narrate incidents of his/her life.

Suppose, the objective of your research is to analyse the gender-based communication patterns in a village, especially the access and use of the Internet but you have little readily available data on which to base your hypothesis on. In this case, you would carry out **exploratory research** in the village, using various primary data tools to collect primary data. Without going into detail because these are discussed in subsequent units, we will now quickly look at various ways of collecting primary data.

Survey

As a first step, you may carry out a household door to door Survey (discussed in detail in unit 7) to determine your sampling frame accurately. This survey would give you information about all the members of each household, their gender, age, and educational level, etc. Such information would help you to select your sample. As such, this would be an exploratory survey. However, if the household survey does not yield enough information to enable you to make your research design; you may consider other methods such as observation or case study.

Observation

For a study on gender-based Internet access and usage cannot be complete until you understand a) if Internet Access is available in the village; and b) where WiFi hotspots are geographically located. For this you would need another tool - **observation**. You would have to walk around the village, see and locate the WiFi hotspots, the cellphone towers, and routers. You may take some photographs and recordings.

You may participate in the village activities in order to get a sense of dynamics. You would then be using a tool called participant observation, also part of ethnographic research. You would then record your observations of the participation in a diary or journal. The diary or journal notes become part of primary data. These tools are part of **Observation** which, in turn, is part of ethnographic research discussed at length in Unit 12 of this course.

You may also go to locations where you see men and women gather and talk to them about Internet Access to gain some insights. By talking to them, you are carrying out **Interviews**, whether with individuals or women in groups. The process of collecting data through Interviews is discussed in detail in Unit 10.

Experiments

A significant amount of primary data can be collected through experiments. Discussed in detail in Unit 9, experimentation is the practice of gathering data by selecting matched groups of people, giving them different treatments or scenarios, controlling related factors in their environments, and checking for differences in their responses. By collecting all this preliminary information in an exploration, you are collecting **primary data**, which will then help you to frame your research questions, hypotheses, sample sizes and samples.

Primary data are not only collected at a preliminary stage. Primary data are also what you will collect during the field-work part of your research. In fact, all the data that are originally collected as part of your field-work are primary data, which will be tabulated, analysed, and reported as part of the research findings.

Both kinds of data, secondary and primary are extremely important and are necessary for research. Without secondary data, one cannot begin to sharpen the focus of research, to define a problem in addressable terms, frame questions and hypotheses, to know what information is necessary to be collected and what is not needed and to make judgments on reliability. When primary data are collected and analysed, findings have to be presented in terms of prior research and evidence collected from secondary sources. Without such a connection, the continuity of the research tradition gets affected.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List three sources of primary data.

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2) Are original interviews primary data?

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- 3) I went to a village and saw that men gather in a tea shop while women gather at the temple. What kind of data have I collected?

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6.5 HOW TO STORE AND SAVE YOUR DATA

At this stage, it is important to know how to store and save your secondary and primary data so that you do not have problems when a) citing sources, and b) doing your research, and c) writing your report.

All secondary and primary data must be stored for at least one year after completion of the research. These data will include your surveys and questionnaires, diaries, interview notes, field notes, observation notes, photographs, audio and videotapes, and anything else you may have collected as part of the research. There are simple reasons for this. First, you might want to consult a source again. Second, you might need to include a bibliography at the end of the research. Third, you need to get the citation correct. Fourth, you may want to write another article or research paper on a different aspect of the project for which the same data can be repackaged. Finally, if you are writing your thesis, the examiner may want information about your source. You should have it readily available for scrutiny when questioned. You may also have to add some more data to strengthen your study, when asked to do so by the examiner, or the agency paying for the research. If you do not take these steps at the beginning itself, you may land yourself in a nightmare later when writing the report, presenting it, or defending it at a viva voce.

In secondary sources, all Internet based sources provide details on how the book or article should be cited in a publication. Follow the rules and methods used in the American Psychological Association (APA) style sheet (<https://www.apastyle.org/>) on how to reference, a book, a journal article, and/or a web page. Save those details either in a folder or file on your computer, or on library cards, with each card for each citation.

All books and reports also provide you with full details on title, authors, publisher, and dates of publication. This information is usually provided in the inside cover page. Note down that information again, in a folder or file so that you can access them readily.

6.6 LET US SUM UP

In this unit, we discussed two types of data collection sources used in research-Primary and Secondary. It was explained that data collected by others, analysed and published in various books, journal articles, reports, etc. constitutes secondary data. Secondary data are very important for research since they help the researcher to set the tone of research, define objectives and questions more sharply, and most important, set the contexts and boundaries within which a research study can be undertaken.

Primary data are what is collected in field, either at a preliminary stage, to provide accurate information on which to build a sampling frame, or to set the field realities. Primary data also form the heart of research, because they are data on which, the findings and conclusions are based. The researcher's primary data becomes the secondary data for the subsequent or later researcher guiding the latter in the way that secondary data guided the study undertaken.

We further examined how we can collect data from Internet sources available for everyone to use. We also explored what are the data sources available in India and how you can use the library effectively and efficiently without much loss of time and effort. You should explore all these sources in doing your research.

Through the discussion we emphasised the importance of data and data sources in communication research because the failure to use proper data sources will affect the quality of your research, as well as the reliability and accuracy. Undertaking research is a time and effort consuming process and there are ways of making it simpler. These ways of sourcing data in a simple way were explained in this unit.

6.7 REFERENCES AND FURTHER READINGS

Chawla, Deepak and Sondhi, Neena (2011), Research Methodology: Concepts and Cases. New Delhi: Vikas Publishing

Hamilton, John (2005), Primary and Secondary Sources. ABDO Publishing.

Singh, Kusum J (1978), Gandhi and Mao as Communicators: A Comparative Study of Practice and Theory. Unpublished doctoral dissertation. University of Pennsylvania 1978

6.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Primary Data are collected fresh and for the first time. Secondary Data have already been collected, computed, analysed and reported by some other person or agency.
- 2) Availability, relevance, accuracy and sufficiency are some factors in choosing secondary data.

Check your Progress 2

- 1) Google Scholar, Social Sciences Citation Index, Arts and Humanities Index.
- 2) JSTOR is a digital library for researchers and students with more than 2500 journals from 57 countries included.
- 3) Two important features of Infilbnet useful for researchers are E-ShodhSindhu and IndCat.

Check Your Progress 3

- 1) Surveys, observations, and interviews.
- 2) Yes, original interviews are primary data.
- 3) Primary data.

UNIT 7 SURVEY METHOD

Structure

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 - 7.7.6 Data Analysis
 - 7.7.7 Evaluation of surveys
- 7.8 Let Us Sum Up
- 7.9 References and Further Readings
- 7.10 Check Your Progress: Possible Answers

7.0 INTRODUCTION

The Survey as a method is one of the most common forms of research methods for collection of primary data from the field. It is a quantitative method and aims to gather data from many respondents to describe the frequency of an event and to generalise results from a smaller sample to larger population depending upon the nature and objectives of a study. Surveys are extensively used to collect and analyse varied types of data including social, political, economic, psychological, technical, cultural and educational among others. However, despite its extensive use, the survey method is not always properly understood or carried out. While using this method, great attention needs to be given on selecting a representative sample, proper designing and pre-testing of questionnaire to reduce error rates, administering, tabulating, and analysing the data.

In this unit, we shall look at the characteristics and types of survey method and its applications. We shall discuss the data collection tools for survey which include questionnaire and interview schedule and examine the process of designing a

questionnaire incorporating both closed and open-ended questions. We shall describe the process of conducting a survey, including planning, sampling, designing and pretesting the research instrument, and analysis of the data collected.

7.1 LEARNING OUTCOMES

After completing this unit, you should be able to:

- discuss the salient features of survey method;
- describe different types of surveys and their applications;
- discuss the relative strengths and limitations of questionnaire and interview schedules as tools of data collection;
- differentiate between closed and open-ended questions;
- design a questionnaire and pre-test it; and
- use survey method in your research design.

7.2 SALIENT FEATURES

The literal meaning of the word ‘survey’ is to take or present a general view of something or examine something carefully. It involves investigation and examination and interviewing people (respondents) and asking them for information. Survey is conducted with a representative sample of the population being studied and it is assumed that information obtained from the sample is valid for the general population. As a quantitative method it describes, explains and predicts.

According to Bertrand & Hughes (2005), surveys can be used for:

- collating descriptive information (television serial viewing habits of homemakers);
- making comparisons (comparing the soap opera viewing habits of teenage girls and teenage boys, or of teenage girls and housewives); and
- exploring relationships (for instance, exploring the relationship between soap opera viewing and level of formal education).

A survey aims to seek diverse opinions and experiences into predetermined response categories and measures reactions of a large number of individuals to a limited set of questions. The method can be used to measure any aspect such as - the number of people below the poverty line, the number of illiterates in a given place, people watching a particular television channel/programme, reading newspaper/listening to radio programme - the list is endless. The numerical data thus obtained is expressed in frequency, percentages, averages, proportions etc. and presented in the form of tables, graphs and charts.

Surveys thus involve numbers, magnitude, and measurement with the focus on objective and standardised means of inquiry. However, the method is not useful for in-depth study of contexts and solutions to problems and it is often argued that ‘Surveys give a big picture of an issue under study but cannot capture the hidden nooks and crannies’.

According to Wiseman & Aron (1970), a survey is a “method for collecting and analysing social data via highly structured and often very detailed interviews and questionnaires in order to obtain information from large numbers of respondents presumed to be representative of a specific population.”

The survey method can be applied for a variety of purposes. It can be used to gather news; business corporations use it to develop marketing strategies; political parties use it to develop campaign strategies; while government agencies use it to obtain data. The scope and depth of a survey may vary depending upon the nature of the study - it may be broad or narrow, it may be limited to a small area or cover an entire state or even several states. Census, opinion polls and exit polls all use survey method for collecting data. In census all the members of a population group are covered thus eliminating the issues of representative sample. However, census requires a huge effort and at the national level can be undertaken only by the government that too only once in ten years in view of the expenses and challenges involved.

7.3 TYPES OF SURVEYS

Surveys can be broadly classified as Descriptive surveys and Analytical or Explanatory surveys.

7.3.1 Descriptive Surveys

In Unit 2 we discussed that descriptive research describes ‘what is’, and involves description, recording, analysis and interpretation of conditions that exist in a given area. Descriptive research uses both quantitative and qualitative research methods to study, describe and interpret what exists at present and provides useful and depth answers to research questions for decision makers and information users. It aims to examine a problematic situation, makes objective observations, analyse and interpret the data in clear and precise terms. Many a time, descriptive research itself is termed survey research. However, it is appropriate to consider survey as a descriptive research method.

Descriptive surveys aim to describe or document the state of affairs as it exists at present - the focus is on present day behaviour. The main characteristic of this type of survey is that the researcher has no control over the variables and s/he can report only of the happenings. Such surveys describe the population being studied and seek to relate this information to their opinions beliefs, values and behaviour etc. To take an example, audience research is conducted to obtain information about the demographic profile such as age, gender, area of residence, historical background, cultural factors, etc., while setting up a radio station. It is important to note that surveys are used to describe a situation at a given point in time; they cannot be used to describe either the past or the future.

7.3.2 Analytic Surveys

Analytical surveys attempt to describe and explain why certain situations exist, why people behave the way they do - what causes certain type of behaviour. These are also called Explanatory surveys as they aim to determine cause and effects relationships between two or more variables. Researchers often use data from descriptive surveys to develop hypotheses and use analytic surveys to test

their hypotheses. The results enable the researchers to examine the inter-relationships between variables and draw inferences.

According to Wimmer & Dominick (2004), a survey has some well defined advantages, because it can be used to investigate problems in realistic settings - where they happen rather than in laboratory or screening room under artificial conditions. The cost of conducting survey is relatively less as compared to other methods which can be further checked by using email and telephonic interviews.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Discuss the main features of Survey Method.

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2) How is Descriptive Survey different from Analytic survey?

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7.4 DATA COLLECTION TOOLS

For conducting Surveys, questionnaire and interview schedules are used as tools of data collection.

7.4.1 Questionnaires

A questionnaire consists of a set of questions posed in a written form and in a definite order. The questionnaires are usually mailed to the respondents, either through regular post or email or administered in person for filling-up the responses. The respondents are expected to go through the questionnaire, understand the questions and reply by writing in the relevant spaces and return the filled-in questionnaires. Since the onus of replying remains on the receiver, it is important that the questionnaire is designed with great care to elicit responses. You will read more about designing of questionnaire in detail in subsection 7.6 of this unit. While mailing the questionnaire through regular post, a self - addressed stamped or pre-paid cover is posted to increase the response rate.

Questionnaires are useful for eliciting data on a variety of areas including demographic factors such as age, gender, family structure, class, education, occupation; media habits - access to media, media exposure and utilisation patterns; likes and dislikes about specific programmes, characters, issues projected in media such as violence, crime and so on.

Strengths

A questionnaire is a convenient and economical tool for collecting large amounts of data from small as well as large population groups. It is an impersonal way to obtain data and ensures a certain degree of anonymity to the respondents; it can be used to obtain personal and to some extent, sensitive information and minimise bias during data collection. It places less pressure on the respondents for immediate response who can respond to the questionnaire at his/her own pace and time.

Limitations

A questionnaire, however, is also beset with a number of limitations - the most important being that it does not produce a high rate of recovery. Studies have revealed that the response rate of questionnaire is rarely more than 12-15 percent and in some cases even less and in such cases, the data has limited validity. Further, a questionnaire can be administered only to those who can read and write reasonably well and cannot be used with small children and illiterates. If a respondent misinterprets a question, there could be inconsistencies in replies. Moreover, respondent may give hurried responses lacking depth resulting in superficiality in responses. They may not reply to all questions and leave some blanks resulting in incomplete data. Once the information has been collected and compiled, analysis of data can be time consuming. Moreover, obtaining representative samples may be difficult leading to sampling errors. All these limitations emphasise the need for taking due care while using this tool of data collection.

7.4.2 Interviews Schedules

Some of the limitations of the questionnaire method can be overcome through Interview schedules or personal schedules. Interview, as you know, is a conversation between the researcher and the respondent - it involves asking questions, listening to individuals and recording their responses. It is generally conducted through face-to-face interaction though it can also be conducted by telephone, emails and other means. A schedule, when sent through post, email etc. is known as questionnaire but when administered in face-to-face situation is called interview schedule or personal schedule as it is a personalised way of data collection. The interview schedule is prepared well in advance before proceeding for data collection. The questions are carefully worded and delivered identically to all respondents and the resulting data are quantitative and comparable.

Strengths

An interview schedule is useful in collecting data from rural and illiterate/semi literate respondents who cannot use questionnaires. Since the interviewer is in a face-to-face situation with the respondents, s/he can build rapport with the respondents, spend time with them and can explain the questions, if needed. The response rate of interview schedule is high as the researcher is in direct contact with the respondents.

It is an effective tool for obtaining information about prevalence and distribution of an issue from large number of people. The schedule once designed can be shown and discussed in advance to weed out inconsistencies or jumps etc. It can be completed quickly in the field and is economical on time. The data obtained

is comparable hence easier to analyse. While less skill is required on the part of the interviewer when collecting data for his/her own research; the field investigators need to be oriented and trained before administering the interview schedule in larger projects involving a number of field investigators for data collection. Field investigators need to understand the broad aim and objectives of the research study and the importance of questions to record complete responses in a proper way. They may need to be trained in how questions should be asked; whether questions can be rephrased; or follow up questions can be asked. They also need to be trained on how to approach a respondent, build rapport, and what to do if the respondent refuses to answer any or all questions.

Limitations

Interview schedules are also beset with some limitations. Much skill is required to create an interview schedule as highly structured questionnaires yield little insight into how people feel about the issues involved. At times, an interviewer may not be able to respond to a valuable issue/situation which emerges during an interview but does not appear on the schedule. Since the researcher is expected to go to the field and meet the respondents to record their replies, it can be time consuming and expensive. It may be difficult to find those selected in the sample, and even if available, respondents may be reluctant to answer some questions. There could be cultural, linguistic and other forms of barriers. Their travel expenses, lodging boarding charges also need to be taken into account. In a larger study, individual differences among field investigators may affect the quality of data hence orientation of the researchers is a must. The tendency to get too personally involved with the respondents at times may turn intrusive and lead to the risk of introducing bias in the results. Care needs to be taken that the researcher does not impose his/her own perspective on an issue. Some of these limitations can be overcome through training of the researchers.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) List five strengths and five limitations of Questionnaire as a tool of data collection.

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- 2) What is the main difference between Questionnaire and Interview schedule?

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7.5 TYPES OF QUESTIONS

Survey questions are largely of two types - Closed-ended or selected response questions and Open-ended or constructed response questions.

7.5.1 Closed-ended questions

In closed-ended questions, the respondents have to select an answer from a list of options provided by the researcher. Such questions are simple to answer, offer uniformity in responses and can be easily quantified; hence they are easier to process as compared to open-ended questions. Open ended questions have been found effective in eliciting data which is sensitive in nature. However, open ended questions are difficult to design as the researcher has to be sure of all the possible range of responses. It is useful for the researcher to have prior discussion with a select group of respondents before designing the questionnaires so that the issues raised by them can be incorporated in the questionnaire to make it more meaningful and effective. Closed ended questions are of following types:

Limited choice questions: which limit possible responses, such as

Do you read newspapers? Yes ☐ No ☐

In some questions the component of Can't say or Don't Know is required. The major disadvantage of closed-ended questions is that often certain responses are not included in the options and to address this, the researcher can include an "Any other" response followed by a blank space to give respondents an opportunity to supply their own answer. Such question allow the possibility of an option given to the respondent

Any other (please specify)

Multiple choice questions: In such questions, the respondents are given the option of ticking many responses. For example, the respondents may be asked:

Which news channels do you like to watch? (Multiple responses)

DD News	AAJTAK	ABP News	India TV
News Express	ETV	Samay	Zee News
News Nation	Mirror Now	NDTV India	India Today
News x	NDTV24x7	Times Now	
CNN IBN	BBC World News	CNN	Other, Please Specify.

However, while formulating such questions, care needs to be taken that the categories are mutually exclusive so that there is no overlap and respondents are informed that they can tick multiple responses.

Ranking scales: Such questions allow measuring the degree of preference to different items, for example:

Which game of sports do you like to watch? (Rank in order of preference in which 1 is the highest)

Cricket	Tennis	Table Tennis	Hockey
Badminton	Kabaddi	Soccer	Hockey
Archery	Athletics (Track)	Boxing	Wresting
Athletics (Field)	Golf	Gymnastics	Rugby
Swimming	Volleyball	Formula 1 Race	Basketball

Any other (pls specify).....

Rating scales: Rating scales allow eliciting varied shades of opinions, beliefs and attitudes on an issue. Likert scale introduced by Renis Likert (1932) is one of the most commonly used ratings scale in which the respondents are asked to choose from the five point scale of Strongly agree, Agree, Neutral, Disagree and Strongly disagree. For example, to ascertain the attitudes and opinions concerning road development, the respondents are asked to mark the extent of their agreement with each of these statements as per scale of:

SA=Strongly Agree A=Agree N=Neutral DA=Disagree SDA=Strongly Disagree

S.No.	Statements	SA	A	N	SD	SDA
1.	Road connectivity is required for development in all the sectors					
2.	Road development is a major link of boosting trade and industries					
3.	Road development should be carried out by the government only					
4.	Private investment for road development is necessary					
5.	People should share investment burden by paying for road usage					
6.	More taxation for road development is needed					
7.	Using private investment and public money together is required for road development					

In this method, answers can be in the form of numerical rating scales for example, 5 for Strongly Agree, 4 for Agree and 1 for Strongly Disagree. An individual's score on a particular attitude scale is the sum of his/her rating on all the items. This enables the researcher to obtain very precise and specific information on attitudes and opinions on a given issue thus add precision to the results.

7.5.2 Open-ended questions

When as a researcher you are not sure of all the possible range of responses and are trying to understand the situation, and you also want to collect qualitative responses; it is useful to include Open-ended questions. As discussed elsewhere, the survey method being a quantitative method is considered too narrow in its approach as it heavily relies on what can be counted and measured. However, there are many aspects of life which cannot be quantified but may hold great importance and open-ended questions allow overcoming this constraint. Open-

ended questions are useful in eliciting data on feelings, beliefs and opinions and help generate more qualitative information even while using a questionnaire. The simplest form of open-ended question relates to the inclusion of ‘why’ and ‘how’ for eliciting qualitative data. Open-ended questions offer freedom to respondents in answering questions and an opportunity to provide in-depth responses. For example,

What kind of change would you like to see in the prime time television news bulletin?.....

How rural news can be included in the prime time television news bulletins?

The limitation, however, is that in such questions, the respondent who is expected to formulate and record answers in his/her own words. The responses require relatively more time in collecting, coding and interpreting. Thus it is useful to have a mix of questions to supplement the quantitative data with qualitative data. Although there is no rule of thumb, four-five closed-ended questions may be followed by an open-ended question.

7.6 DESIGNING A QUESTIONNAIRE

It is often said that your study is as good or bad as your questionnaire, hence great deal of care needs to be taken while designing a questionnaire. A well designed questionnaire elicits quality responses. A set of questions are framed in the questionnaires and the researcher mails them to the respondents with a request to return them after completion. It should be easy to understand, appealing and motivating to the respondents.

It is useful to include a short introduction explaining the purpose of the study and the expectations from the participants. Informed consent needs to be taken and they need to be reassured that their participation is voluntary and their answers will be kept confidential and used only for research purpose. Their names will not be mentioned and they should feel comfortable in sharing the information. Towards the end, the participants need to be thanked for their participation and researcher may share his/her contact address for any further information.

The questionnaire needs to be structured keeping in view the objectives of the study. For example, if the objectives of your study are: ‘To obtain viewer’s feedback on the news content and presentation aspects of Prime time news bulletins’ some of the following areas may be considered:

- Personal Profile Media access
- Media utilisation patterns
- Content of primetime news bulletins Presentation of the news stories

The questions related to personal profile of the respondents are generally given at the beginning in which information pertaining to socio-demographic profile of the respondents such as age, gender, religion, caste, area of residence, type of household, type of settlement, educational qualifications, occupation, income levels is elicited. It is important to include only what is necessary in the light of your objectives and the temptation to cover every possible area in the subject needs to be resisted. Many a time, researchers try to cover extensive details of an issue under study but the questionnaire should not be extremely lengthy. The

respondents may not have time and inclination to respond to a very detailed questionnaire.

Further, given the limitations of survey method, all the issues under study cannot be examined through questionnaire and different methods of data collection will have to be used - for example, some issues will have to be analysed through case study, observation method or in-depth interviews. The bottom line is that each question should have a purpose. A well structured questionnaire helps at the time of coding of data and analysis as well. Informal interviews in the study area have been found useful to identify, develop and refine the questions.

While designing the schedule, the flow and sequence of questions needs to be checked - early questions should be easy to answer. It is always useful to have a variety of questions as discussed in the previous section depending upon the nature of the study. Many a time people tend to give politically correct answers leading to a skew in the findings hence it is important to include some questions to cross-check their responses. However, only relevant questions which meet the objectives of the study should be included - unnecessary questions only add fluff, hence better avoided.

If a questionnaire has to be administered in Hindi or any regional language, then either it should be developed in the same language or translated in the language to be administered. This is important as every time a question is translated, the field researcher is likely to add or omit some information which may be understood differently by diverse set of respondents leading to disparity in responses. Space in questionnaire is also an important issue which needs to be conserved as you will have to get many sets of the questionnaire be made.

Certain do's and don'ts need to be kept in view while designing a questionnaire.

7.6.1 Do's

- At the outset state concisely in a note what the survey is all about and how the findings will be used. Request the respondents for their cooperation.
- Write questions in simple language which is easy to understand.
- Keep sentences short and to the point.
- Keep the level of education, knowledge and perception of the respondents in view while framing the questionnaire.
- Begin with demographic profile (age, gender, education, area of residence, occupation, income level etc.). May include psychographic profile, if needed.
- Proceed in a logical order-from simple to complex; general to specific.
- Ask only one piece of information per question.
- Define the term for the first time to remove ambiguity e.g. Have you heard of euthanasia (mercy killing).
- Check that nothing important has been left out.

7.6.2 Don'ts

- Include 'leading' or 'loaded' questions which lead the respondent to answer in a certain direction (Do you think drinking alcohol has become a status

symbol and those who don't drink become a centre of embarrassment?).

- Include double barrel questions (Is media glorifying drinking and leading the youth in drinking?).
- Include ambiguous questions as they will yield ambiguous data (Are multiple classes being handled by one teacher in the school?).
- Include questions which require unreasonable effort to answer.
- Include questions that cannot be answered accurately.
- Ask embarrassing questions.
- Ask questions which are not relevant to the study objectives as it can generate enormous data which may be later difficult to organise and analyse.

It is also important to remember that the questions asked on a questionnaire or in an interview must be closely related to the research questions and hypotheses framed for the study. There is often a tendency to collect too much data and then not know how to connect the data to the research questions or hypotheses. Thus, every question must be connected to the objectives of the research being undertaken.

Activity 1

Design a questionnaire comprising 15-20 questions on a topic of your choice keeping in view the points discussed in this sub-section.

7.7 THE PROCESS

The following steps are involved in survey method:

- Planning
- Sampling
- Designing research instrument
- Pre-testing
- Administering the questionnaire
- Data Analysis

7.7.1 Planning

The planning stage of a survey involves identification of topic and defining the research problem. As discussed in Unit 3, Block 1, you need to consider what you will be investigating and why this research is worth investigating. It is also useful to consider at this stage whether survey will be an appropriate research method to meet the objectives of the study, the population to be studied, study area, hypotheses for testing, data analysis techniques to be used and so on.

7.7.2 Sampling

The next stage is making decisions about the sampling frame, sample size of the population that need to be included in the survey. We have discussed Sampling in detail Unit 4. Remember, a reliable sample must be representative and adequate in size. If the sample is not representative of the population being studied, the findings may not be valid and reliable.

Conducting large surveys involves a great deal of coordination, training and supervision. Large surveys are also expensive. Language also places huge challenge apart from sample selection. A normal sample size for a state level survey is kept around 3500-4000, while in national studies; the sample size is around 15000 with at least six big states being included. The heterogeneity of population even within a state means that ideally every constituency and significant social groups within each constituency need to be sampled. What is important to remember is that it is the representativeness of the sample, not necessarily the sample size that makes survey results reliable.

7.7.3 Designing the research instrument

You need to keep the issues raised in sub-section 7.6 in view while designing a questionnaire. Once your draft is ready it is useful to discuss it with your seniors or colleagues for their inputs for finalising it.

7.7.4 Pre-Testing of Questionnaire

Testing the draft questionnaire on a small segment of the potential survey population is called pre-testing. This is an important phase of using surveys, as testing enables you to identify the gaps in the questionnaire, if any. It helps to ascertain whether respondents can easily understand the questions or not. It helps to remove ambiguity and ensure clarity. It answers questions such as - are there any missing links; does the questionnaire elicit the information required and does it meet the objectives of the study? Is the sequencing of the questions proper or is there a need for taking a different approach? Based on the feedback, the questionnaire is modified and firmed up. Thus pre-testing helps you as researcher to identify problems and solve them before proceeding to undertake a larger study.

7.7.5 Administering the Questionnaire

Once the questionnaire has been pretested and firmed up, it has to be administered to the respondents - either through post, telephone, email or in person. However, in view of the low response rate, as discussed earlier, there will be still a need to follow up by sending a polite reminder and/or attaching another copy of the questionnaire. If using personal interview schedule, you will have to introduce yourself and share the purpose of the study and seek cooperation. While filling up the schedule care should be taken so that you do not to give your own opinions, even by mistake or accident.

7.7.6 Data Analysis

Data analysis is the processing of collecting raw data and converting it into information useful for the stakeholders. The data is analysed to answer questions, test hypotheses or disprove theories. Data analysis involves processing of data, interpreting the results and drawing findings.

For data analysis a variety of software are available which may be used in view of the nature and objectives of the research study. We have discussed data analysis techniques in Unit 15, Block 4 in detail stating that data analysis through statistical procedures represents a systematic and objective way of determining whether significant patterns of relationships exist among those phenomena that have been measured in data collection.

Findings are drawn from the analysis of the data. Interpretations of the findings are provided by drawing inferences, linking the findings with other studies - how these are similar or different from existing research. It is explained as to what new element has been added through the findings and how this is important for understanding a certain phenomenon.

There is a need to eliminate bias or error and remove inconsistency as well as contradictions in findings to increase validity of the research findings. These findings are presented in tabular forms and represented in charts and graphs - these could be represented in either pie chart, bar diagram, line diagram and so on. Findings and your experience are shared with others to add to the scholarship in the discipline and theory and practice of life.

7.7.7 Evaluation of Surveys

According to Berger, there are three considerations involved in evaluating surveys:

- Sample size
- Margin of error
- Confidence level

The error of measurement strongly depends on the sample type and size. For example, to study the media habits of a village, if only the upper classes and castes or males are included in the sample, the results will be skewed. The margin of error decreases as the sample size gets larger but in smaller increments.

According to Berger, the larger the sample, the more confidence you can have that your findings will be accurate, but this works only up to a point as large samples may not provide the equivalent amount of accuracy and a survey of 1,500 people will give as much accuracy as a survey of 2000 people.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Why is it important to draw a representative sample?

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2) List the advantages of pre-testing.

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3) What are important factors for evaluating surveys?

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7.8 LET US SUM UP

In this unit, we discussed various issues related to Survey method. It was discussed that the concept of Survey originally developed in social sciences is one of the most commonly used research methods in mass communication as well. Survey method is an important tool of developing audience profile, exploring media habits, and media exposure and utilisation patterns among others. It aims to measure and count the frequency of an occurrence systematically and objectively. It is basically a quantitative technique but can also elicit qualitative data in a limited way through the use of open-ended questions.

We further discussed the salient features and characteristics of survey method, the types of survey, the need and importance of designing a good questionnaire, the process of undertaking a survey. It is hoped that the learning acquired will enable you to use this method in your own research study in a systematic way.

7.9 REFERENCES AND FURTHER READINGS

Berger, Arther Asa (2000), Media and Communication Research Methods, Sage Publications, London, New Delhi.

Bertrand Ina & Peter Hughes (2005), Media Research Methods: Audiences, Institutions, Texts, Palgrave MacMillan, New York.

Priest S. H (2010), Doing Media Research: An Introduction, Sage Publications, New Delhi.

Wimmer R.D. & Joseph R. Dominick (2004), Mass Media Research: An Introduction, 4th edition, California, Wadsworth Series in Mass Communication.

7.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Survey is a quantitative method of data collection which is used to describe the frequency of an event and generalise results from a representative sample to larger population. It seeks diverse opinions and experiences into pre-determined response categories and measures reactions of a large number of individuals to a limited set of questions. It is assumed that information obtained from the sample is valid for the general population.
- 2) The focus of Descriptive survey is to describe present day behaviour whereas Analytical survey attempts to explain why certain situations exist, why people behave the way they do and what causes certain type of behaviour. Often

data from descriptive surveys is used to develop hypotheses and analytical surveys relate this information to analyse the opinions beliefs, values and behaviour etc. of the respondents.

Check Your Progress 2

- 1) Five strengths of questionnaire are:
 - Useful for eliciting data on a variety of areas, such as demographic profile, media habits, issues projected in media such as violence, crimes, etc.
 - Convenient and economical tool for collecting large amounts of data from population.
 - Impersonal way to obtain data hence ensures certain degree of anonymity to the respondents.
 - Can be used to obtain personal and to some extent, sensitive information.
 - Places less pressure on the respondents for immediate response who can respond to the questionnaire at his/her own pace and time.
- 2) Five limitations of questionnaire are:
 - The response rate of questionnaire could be as low as 12-15 percent.
 - Can be administered only to those who can read and write reasonably well.
 - Designing a questionnaire requires great skill on the part of the researcher.
 - If a respondent misinterprets a question, there could be inconsistencies in replies.
 - The responses may be lacking in depth resulting in superficial data.
 - They may not reply to all questions and leave some blanks resulting in incomplete data.

A questionnaire is administered through email, by post or in person to the respondents and has to be filled-in by them. In contrast, the interview schedule has to be administered by the researcher to the respondents in a face-to-face situation. It is a personalised way of data collection therefore is also called personal schedule.

Check Your Progress 3

- 1) It is important to draw a sample which is representative of the population being studied for ensuring reliability and validity of the findings.
- 2) Some advantages of pre-testing are:

Pre-testing enables the researcher to identify the gaps in the questionnaire. It helps to remove ambiguity and ensure clarity.

It ensures that the questionnaire meets the objectives of the study.

It helps the researcher to solve the problems before proceeding to undertake a larger study.
- 3) Three main considerations involved in evaluating surveys are - sample size; margin of error and confidence level.

UNIT 8 CONTENT ANALYSIS

Structure

- 8.0 Introduction
- 8.1 Learning Outcomes
- 8.2 Conceptual Foundations
 - 8.2.1 Concept and Definitions
 - 8.2.2 Origin and Growth
- 8.3 Characteristics of Content Analysis
 - 8.3.1 Applications
 - 8.3.2 Limitations
- 8.4 Types of Content Analysis
- 8.5 Process of Content Analysis
- 8.6 Let Us Sum Up
- 8.7 References and Further Readings
- 8.8 Check Your Progress: Possible Answers

8.0 INTRODUCTION

Communication as you are aware is basic to human existence. Initially it was visual and verbal in nature but with the development of the written script and digital encryptions, these too became a part of how we humans communicate with each other and with our own selves. Language has always been very complex, and different people, as per their culture, belief systems etc. may encode and decode (give meaning to and derive meaning from) messages differently. In many instances, the medium also brings its own grammar - for example, in audio visual media as also in computer mediated communication, messages are structured differently, leading to different meanings in different contexts.

In the light of above, it becomes important to construct and deconstruct communication content in order to understand how we evolve as individuals and as a society.

In this unit, we shall study content analysis in terms of its growth and development as a research method. We will focus on its uses and limitations in understanding what is communicated, the manner in which it is communicated and the projected effects of the communications on human and social behaviour. We will also focus on the location of meaning in communication content and how various types of content analysis method are used to describe meaning as it lies on the surface as well as when it serves the undercurrents of a communicated message.

8.1 LEARNING OUTCOMES

After working through this unit you should be able to:

- discuss the conceptual foundations of content analysis;
- describe the characteristics of content analysis;
- explain its types and applications;

- differentiate between quantitative and qualitative approaches of content analysis; and
- use content analysis method in your research work.

8.2 CONCEPTUAL FOUNDATIONS

In this section we will study the concept behind content analysis as a research method. We will also trace its growth trajectory driven by various socio-political factors. This will give you an insight into the uses and limitations of this method.

8.2.1 Concept and Definitions

There are different ways of defining content analysis. In the definitions listed below you will see a variation primarily when it comes to the location of meaning in the text. A definition that stresses on the *manifest* or visible content is usually one that follows a quantitative approach to content analysis as it looks at surface meaning. A definition that stresses on *latent* or invisible content is one that follows a more qualitative approach to content analysis by searching for meaning beneath the surface.

The most classic definition of Content Analysis is provided by Bernard Berelson, also known as the father of Quantitative Content Analysis, in his book, *Content Analysis in Communication Research*: “content analysis is a research technique for the objective, systematic and quantitative description of the manifest content of communication” (1952:18).

Kerlinger’s definition involves four concepts that require further elaboration. Firstly that content analysis is objective i.e. the researcher’s personal biases and idiosyncrasies should not enter into the findings. Operational definitions and rules for framing content categories, defining units of analysis etc. should be comprehensive and explicit so that another researcher could replicate the study and come up with similar findings. Secondly, content analysis should be systematic i.e. there should be consistent application of procedure viz. sample selection, data evaluation etc. Thirdly, content analysis should enable a precise representation. For instance, the statement: ‘75% of all primetime programmes contain at least one act of violence’ is more precise than ‘Most programmes are somewhat violent’. The quantification also gives researchers statistical data that can be used for further interpretation and analysis. Fourthly, Kerlinger refers to the manifest content of communication. This is where he primarily differs from those favouring a more qualitative approach to content analysis. He refers to the content that can be seen/read and heard i.e. the manifest content that is easily observable and measurable.

Klaus Krippendorff, in his book, ‘Content Analysis: An Introduction to its Methodology’, takes a more qualitative approach. He defines content analysis as a “research technique for making replicable and valid inferences from data to their context” (1980: 21). The focus here is more on the symbolic meaning of messages which implies that any single written message can have multiple psychological, sociological or political interpretations and all of these may be simultaneously valid. We receive these messages from someone who framed them as per his/her context and we try to make inferences about the context by trying to interpret the message in a valid manner. For instance, a content analysis

of the news coverage of a riot might conclude that 90% of the coverage showed non-violent scenes. But the 10% that contained violence might have been so powerful that its impact on the audience was far greater than the non-violent coverage.

Krippendorff evolved a conceptual framework within which content analysis is carried out: the data as it is communicated to the analyst; the context of the data; the analyst's own knowledge; the target of a content analysis; inference as the basic objective; validity as the ultimate criteria. A TV soap opera is communicated to you through words and images; as a student of communication you decide that you will analyse it only from the sociological context; you then bring your knowledge of culture, ethos, denotative and connotative meanings into the analysis; you also specify your objective in carrying out the analysis, i.e. message analysis in terms of sender-receiver interactions. You could also hope to draw inferences from the data obtained, and hypothesise an association between 'reel' reality and 'real' reality.

Charles R. Wright in his book *Mass Communication: A Sociological Perspective* adopts a more balanced approach: "content analysis is a research technique for the systematic classification and description of communication content according to certain usually pre-determined categories. It may involve quantitative or qualitative analysis or both. Technical objectivity requires that the categories of classification and analysis be clearly and operationally defined so that the other researchers can follow them reliably (1986: 125). However, it is important to remember that content analysis itself cannot be used to determine anything about the communicator, or about the impact of the communication on audiences for which other research methods need to be used.

8.2.2 Origin and Growth

The term *content analysis* is over half a century old. It was first listed in the Webster's Dictionary in 1961 but its intellectual roots go far back, in fact to man's first consciousness of language and symbols. We find it important not only to analyse written scripts - in some cultures the letters of the alphabet are pictograms and ideograms in themselves- but also the tone of the language used i.e. the paralanguage, if we want to grasp the content.

Early Content Analysis: Empirical inquiries into communications content date back to the 1860s when the Church began to be increasingly concerned with 'non-religious' content as carried by the newspapers of the day. More general studies have been documented in the late 19th century. For instance, G.J. Speed published a study in 1893 that documented how religious, scientific and literary matters had dropped out of leading New York newspapers between 1881 and 1893 in favour of gossip, sports and scandals. As other mass media - radio, cinema, and television (and now the Internet) - came into being, this approach began to be used to study both the manifest and the latent content of communication, as carried through these channels.

What we need to understand here is that these media are technology-driven, more powerful, more sensory, and replete with both verbal and non-verbal cues. Moreover, they are also part of a socio-political environment that is more complex, demanding its own measures of analysis. Studies started being conducted on opinion formation through mass media; the pro-con attitudes held by media have

been evaluated to make predictions regarding social behaviour and formation of stereotypes; the issue of agenda setting comes to the fore in propaganda analysis; comparative studies are conducted on how history was written and recorded for the same event as interpreted by diverse value sets.

Propaganda Analysis: The Second World War, in particular, provided a major impetus for the use of content analysis as a research method in all areas of social sciences - sociology, psychology, economics, and literature, among others. For instance, psychologists in Allied Military Intelligence Units monitored the number and types of popular songs played on European radio stations to determine the troop movement of the Nazi forces. Religious and political speeches were analysed to determine the use of propaganda tools such as glittering generalities, bandwagon, name calling, card stacking etc. and how these were used to command obedience and generate patriotism - a technique referred to as 'Preparatory Propaganda'. Speeches were also analysed to determine personality characteristics of the speaker, particularly those engaged in political battles.

Generalised Content Analysis: As human history became more and more complex, anthropologists started using content analysis to study myths, folk tales and fiction to determine the changing trends in social structure and relationships. Historians used it to establish authorship of unsigned documents - writing patterns in documents of established authorship were matched to the doubtful ones in order to determine whether the same author had written both. As knowledge generation became more and more important, educational material began to be recognised as a rich source of data both to make inferences about the processes of reading and to understand larger political, attitudinal and value trends to be found in the textbooks.

Computers in Content Analysis: The large volume of documents to be analysed and the repetitiveness of the task saw researchers turn to computers in the late 1950s for mechanical translation, mechanical abstracting and information retrieval. Computational Stylistics emerged as a new discipline for literal (as opposed to numerical) data processing. Linguistics developed numerous approaches to analysing sentence structure and word semantics. For instance, it was determined that out of all the choices available, why were only certain words selected and when. This way the politics of knowledge began to be studied. Today, we talk of Artificial Intelligence wherein robots are programmed to understand natural language. The Internet too is moving into its third phase - the Semantic Web. Computers are now programmed to understand the meaning and the context of word/voice/thought commands and process action accordingly.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Define Quantitative Content Analysis.

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2) Who is considered to be the father of Quantitative Content Analysis?

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3) When were the first studies using the content analysis method documented?

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8.3 CHARACTERISTICS OF CONTENT ANALYSIS

Even though journalism and mass communication is seen as the historical origin of this technique, content analysis is applied in various disciplines ranging from psychology to anthropology. Studies range from mere description to determining associations and causal relationships.

8.3.1 Applications

The primary applications of content analysis are as follows:

- i) **Describing communication content:** Content analysis describes what exists i.e. it describes the characteristics of a particular communication content by stating the what, how and to whom something is said in it. For example, how live reports are used in TV news bulletins or what is the gender depiction in soap operas. These studies can also be used to study societal change. For instance, changing public opinion on current issues can be determined by analysing newspaper editorials over a period of time or trends in consumerism can be studied through a historical content analysis of advertisements. Studies are also conducted on readability of content, use of stylistic devices, text to visual ratios etc.
- ii) **Making inferences regarding message characteristics:** Content analysis can be used to determine the association between source and message. For instance, a comparative analysis can be done of how the same event is covered by two different publications - the hypothesis being that if the source has characteristic A, then messages containing elements x and y will be produced, while if the source has characteristic B, then messages containing elements w and z will be produced. Here, inferences are drawn as to why something is said as also what, where, when, how, which etc.
- iii) **Comparing media content to the 'real' world:** Content analysis is often used in comparative studies where a situation, event, phenomenon existing in reality is compared with its media presentation or representation. For instance, cinematic depiction of certain ethnic communities as law breakers as compared to actual data; or the role that a woman plays in most advertisements as compared to the roles she performs in real life.

- iv) **Assessing the image of population groups:** The manner in which certain minority or dominant groups are represented in media is analysed to determine under-representation/over-representation or misrepresentation. For instance, whether all Indian ethnicities are represented equally in mainstream media or whether the marginalised continue to be fringe elements? Whether women are only asked their opinion in their capacity as housewives rather than be invited to TV studios as experts on economics, polity and so on.
- v) **Establishing a starting point for Effects Studies:** Here content analysis is used as a part of a two-pronged study. First the dominant message /theme of communication content is determined. Then a separate study is conducted to determine whether the content had any effect on those who consumed it, especially the heavy users. For instance, whether heavy viewers of violent TV programming tend to be more fearful of the world around them since it cultivates attitudes more consistent with its messages than with reality. Content analysis is also used in studies on agenda setting where the objective is to determine how personal agendas become public agendas and how various propaganda techniques are used for the same.

8.3.2 Limitations

Content analysis as a research method has some intrinsic limitations as well. The nature and degree of limitations depends upon the approach used to carry out a content analysis.

- i) Content analysis alone cannot serve as the basis for claims about media effects, for instance, just by doing a content analysis of editorials you cannot determine that a newspaper's readers also conform to the same viewpoint. You will have to conduct a readership survey or a text-reception analysis in order to determine whether the text was decoded the way it was encoded. You will also have to do a source analysis in order to determine what was coded.
- ii) The findings of a content analysis study will depend upon the content categories that were adopted as well as their operational definitions. For instance violence as a concept is extremely complex, a study may define violence in physical terms and analyse a film accordingly. Another study may analyse violence in psychological terms and the same film will be analysed differently and arrive at different conclusions.
- iii) Certain types of studies are difficult to conduct since there might not be content pertaining to them or the level of incidence may be very low. Though this in itself is a significant finding, it makes the job tougher for a researcher. For example, there are many reclusive tribes in India which are barely covered by media or in popular literature. Therefore it becomes difficult to undertake a study on their projection / representation outside of reference texts. Many a times, a large volume of media content needs to be studied in order to conduct an analysis. For instance, a small sample, may be 12 issues a year, is enough to determine how political stories are covered in print media but it may be necessary to take up to 24 issues if the coverage of development issues is to be analysed since their coverage will not be so frequent.

- iv) Conducting a content analysis can be time consuming and tedious since it can involve analysis of a large volume of data. There can also be problems regarding the sourcing of data, particularly archival material or classified documents. Further, if the material is in electronic form, then there can be the additional expense of videotaping, playback, transcription, etc.
- v) Analysis can also be a painstaking process particularly if the qualitative approach is adopted and the meaning is located beneath the surface, a great deal of data mining is required here. In order to ensure scientific rigour, it also becomes necessary to go in for investigator triangulation i.e. where more than one researcher from different backgrounds, analyse the content in order to arrive at a consensus. In quantitative content analysis it becomes important to test for intra-coder and inter-coder reliability so that each content unit is coded in its relevant category.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List two uses of the content analysis method.

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2) List two limitations of the content analysis method.

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8.4 TYPES OF CONTENT ANALYSIS

As mentioned earlier, there can be two approaches to doing a content analysis - **the quantitative and the qualitative approach**. The basic difference between the two is as regards the location of meaning and the manner in which inferences are drawn. The former emphasises a fixed meaning that is on the surface i.e. easily observable and measureable. It can be repeatedly identifiable by different researchers using the same analytical framework. The latter searches for multiple meanings, usually found beneath the surface meaning - they can drive it or even contradict it. Moreover, this approach focuses as much on source and reception analysis as it does on message analysis. Therefore, the qualitative approach does not lend itself to easy replication.

In practice we find both approaches being used. Through quantitative techniques we establish the frequency of occurrence or the comparative location of different categories of content and through qualitative techniques we understand the meaning of its message.

Qualitative Content Analysis

The types discussed below are qualitative content analysis as these are derivatives of the basic quantitative content analysis.

Bias Analysis

By defining categories of direction in opinion (for, against, or neutral), it is possible to determine the slant or bias in coverage. Slant or bias is studied using similar categorisation as for quantitative analysis. However, often the unit of analysis is the whole item or story instead of the word, space or time. This is because slant or bias depends on contexts much more than on what is printed or visible. The difference is that coders are asked to determine the bias as “pro” or ‘against’ or as ‘neutral’ By analysing content in this manner, it is possible to infer whether the content of the newspaper, television, programme, etc, has a hidden bias in terms of its coverage.

Semiotic Analysis

This type of analysis is primarily concerned with how meanings are generated. The content is divided into two components - the signifier and the signified - and meaning is derived from the relationship between the two. For instance, the use of rockets, gizmos, extra-terrestrials etc. signifies the future. Or the use of candlelight and roses signifies romance. Meaning is also understood in terms of what is not signified. In the case of both the examples it is also simultaneously signified that these are not commonplace or routine situations.

Meaning is not fixed in semiotic analysis - it has many cultural, ideological and linguistic associations. For example, we immediately associate Mahatma Gandhi's attributes with someone wearing round wired spectacles. Similarly, the use of different colours during a wedding - primarily white in Western countries, and primarily vivid colours in India signify cultural differences with different meanings. You will read in detail about Semiotic analysis in Unit 13 of this course.

Discourse Analysis

In this analysis, every situation that is talked /spoken/written about is looked at through two models - the event model i.e. the situation itself; and the context model - the manner in which the source sees it and the receiver receives it. Here we speak of the politics of knowledge. When a journalistic reports on an event, may be a political rally, s/he has many words at disposal but s/he makes a conscious choice. This choice is not random or accidental but deliberate since s/he wants to convey a particular meaning. S/he can use the word ‘crowd’ which is neutral or use the word ‘mob’ which has negative connotations. The choice of words will dictate how the reader will receive the meaning though the reader too will bring in his/her own perceptions while decoding the news report.

Rhetorical Analysis

Rhetorical analysis is to do with the study of persuasion and is most often carried out in the study of advertising and other forms of public communication. In this, we focus on the character and credibility of the source (ethos); the use of emotional appeal (pathos); the use of rational appeal (logos); the purpose (aim) of the discourse and the medium (mode) used.

In this type of analysis, the manner in which the content is structured is studied. A news report for instance is broadly divided into the summary (which contains the headline and the lead paragraph) and the background (which contains the main events, the context and the history). Here the analysis is done on the basis of what is included/ excluded as also the extent of detailing provided.

Narrative Analysis

In this type of analysis we study how the story is narrated within the formal structure. Meaning is generated through the manner in which the characters are handled and the plot unravels. Vladimir Propp, in his 1958 study, 'Morphology of the Folktale' identified 31 functions and 7 agents through which the story unfolds. According to him every folktale has a hero, heroine, villain, helper etc. and all go through situations such as failure, victory, deception, search etc. It has been seen that almost all films, plays and novels have most of these situations and agents.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) What is the basic difference between quantitative and qualitative content analysis?

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- 2) How is meaning determined through Semiotic Analysis?

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- 3) Ethos, pathos, logos etc. are areas of study under which type of analysis?

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8.5 PROCESS OF CONTENT ANALYSIS

The basic steps in carrying out either a quantitative or a qualitative content analysis are listed below.

- a) **Defining the research problem and formulating the terms of enquiry:** As with all research, it is important that you select a research problem that has some significance, whether it serves to increase knowledge or whether it solves a practical problem. Doing a review of literature or your own observations can help define a research problem. It can be based on existing theory or it can be a response to a current or potential social situation. A content analysis can be conducted to determine whether the growing acceptance of homosexuality or live-in marriages in society has found equal acceptance in our films. Or whether there is any difference in the content of state-owned media as compared to private ownership. The enquiry can be limited to just description or you can go a step further and also seek to analyse the nature and degree of associations/correlations. In the latter case, the data obtained would have more utility.
- b) **Defining the population and selecting the sample:** When doing a content analysis, the content source and the time period usually dictate the boundaries of the population. For instance, if the research problem is to study the portrayal of women in TV soap operas, we first define what we mean by TV and by soap operas. For the purpose of the study we may define TV as only entertainment channels and we may define soap operas as any fictionalised programmes with a continuous story line. We may further limit the population to only prime time soap operas appearing on a specific television channel, in a given year.

As mentioned earlier, we can sample by using content source and time period as our criteria. For instance, we can make a list of all the prime time soap operas and choose a requisite sample by using either a probability or a non-probability sampling procedure (refer to the sampling methods discussed in detail in Unit 4). We can go in for simple random sampling and select a certain number of soap operas at random from the list or we can choose every *n*th programme on the list by going in for systematic random sampling. If one of our research hypotheses is that different production houses have different policies regarding the manner in which women are depicted, then we can first stratify the list in terms of production houses and then select a sample of programmes from each stratum.

We can follow the same multi-stage procedure when working with print media. For instance, if our research problem is to do trends in political news coverage, we can first define the population by focusing on only daily newspapers, may be only those published in North India. We can further limit our study to only front page stories. If we have a hypothesis that a newspaper's circulation influences the manner in which it covers the news, then we can stratify the population on the basis of circulation and then select newspapers from each stratum. We can also sample as per time period by selecting one issue per month. Or we can select the first Monday issue of January, the first Tuesday issue of February and so on. The idea is to put a system in place for sample selection.

- c) **Constructing content categories:** There are three factors that are of extreme importance in this particular step - **the categories that are constructed should be exhaustive, mutually exclusive and reliable.** The entire validity of the research depends on this step. If the research purpose is to determine the kind of content carried by a newspaper, then we can have the categories

of news reports, features, special articles, editorials, as well as a miscellaneous category. If our categories are exhaustive then most of the content will be adjusted in the first four categories itself. But if most of the content falls in the miscellaneous category then that means that there is scope for more categories to be defined.

The second important factor is that of exclusivity. The categories should be defined so clearly that the coders should have no problem in identifying a particular content as belonging to a particular category alone. For e.g. a content containing only facts relating to the 5 Ws and 1 H would be identified as a news report while a lengthy opinion piece carrying a by-line would be identified as a special article.

- d) **Inter-coder Reliability:** It is essential that a pilot study or pre-testing be conducted to ensure both inter-coder and intra-coder reliability. This stage is also called the coder-reliability test. In the first case, two different coders analyse the same content and their degree of agreement is determined. If both code the content in the same category or do so in a majority of instances, then that means that the categories were defined clearly. In the latter case, the same coder analyses the same content after a time gap. If his/her coding the second time around is the same or nearly same as the way s/he had done it the first time, that means that the definitions have reliability. To get around two coders analysing the content differently, an odd number of coders can be taken, i.e. 3, 5, or 7. In this case, the majority opinion holds.
- e) **Selecting and defining a unit of analysis:** The unit of analysis constitutes what is to be counted for the analysis. In written communication it could be a word, a sentence, a paragraph, an entire article/story. Written communication can also be analysed in terms of theme. Or in terms of format e.g. dialogue versus description. For instance, if the research objective is to determine the manner of language used, then the colloquial use of words as compared to pedantic usage can be counted in a given text. The unit of analysis can also be in terms of print space. For e.g., a ruler can be used to determine how much square column inches does a newspaper devote to event-based stories as compared to issue-based stories.

In analysing television and cinematic content, the unit of analysis can be characters, acts, entire sequences or programmes. The unit of analysis can also be in terms of time. For instance, to determine the extent of violence in a programme, the unit of analysis can be defined as a single physical act resulting in bloodshed. These acts can then be counted or timed. It is very important to clearly define the units of analysis. In the case of this example, if it is not clearly defined that a unit is one physical act, then different researchers may come up with different findings while replicating your study. One researcher may consider every blow to be one act while another researcher may consider a continuous exchange of blows to be one act. It is essential that a pilot study be conducted - especially if multiple coders are being used for the study - in order to ensure that there is complete clarity. Careful definition of the unit of analysis is essential to avoid confusion at a later stage.

- f) **Coding the content:** After the content categories have been determined as also what is to be counted, the coding or actual counting is done. This can

involve all four levels of data measurement. In the nominal level we simply count the frequency of occurrence of the units in each category. To take up the previous example, we can simply count the number of news reports, features, special articles and editorials that appear in the sample issues of a newspaper. Or we can count the number of times that different sales appeals are used in advertisements constituting the sample. Ordinal and Interval level of measurements are used when we want to rate or rank content attributes.

Here the numbers that we assign have numerical significance unlike in the nominal level. For instance, we can identify and rank different positions on the newspaper page in order of importance and then determine what kind of content appears in the most important position and what kind of content is relegated to the least important position on the page. In the Interval level we can use scales such as the Semantic Differential Scale where we use anchoring opposites usually separated by seven equidistant points. For e.g. when analysing the portrayal of women in soap operas, we can rate them as being shown as submissive - aggressive, traditional - cosmopolitan, weak - strong and so on. A rating on the 7-point scale is given as per the strength of the attribute possessed - a rating of one to three would veer more towards the negative trait while a rating of above four would veer more towards the stronger trait.

At the ratio level, we usually analyse content in terms of space and time. For example, we can determine the verbal-visual ratio by measuring the square column inches devoted to text as compared to the space devoted to pictures and other illustrative material in a publication. We can also compare how much time is devoted to social issues in a news bulletin as compared to political issues or how much time is devoted to negative stories such as crime, corruption etc. as compared to positive stories on scientific breakthroughs, economic development, administrative efficiency etc.

The coding can be done manually or by using computer software. The latter is more reliable when the categorisation system is unambiguous. Popular software for text documents are Text Smart, NUDIST, Profiler Plus etc. while programmes such as Camera and Executive Producer can be used to code audio-visual content. The data can also be tabulated into coding sheets such as Excel spreadsheets.

- g) Drawing conclusions and inferences:** Data analysis is extensively discussed in Unit 15 of this course. However, briefly, it can be said that if the quantitative approach is used in content analysis, then conclusions regarding the data can be drawn using both descriptive and inferential statistics. We can use percentages, means, deviations etc. as also statistical tests such as chi square, t-test, ANOVA etc. to test hypothesis pertaining to associations, correlations etc.

Check Your Progress: 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) What are the three criteria for a good categorisation system?

2) Name the different levels of measurement.

3) What is the most frequently used sampling method in content analysis?

8.6 LET US SUM UP

In this unit we discussed the definitions of content analysis, both from the quantitative and the qualitative perspectives. We traced its growth and linkage with important socio-political developments including the World Wars. We also analysed the uses and limitations of this method and then classified it into two broad categories: Quantitative and Qualitative content analysis - each sub-type locating and understanding the meaning of the message from different perspectives.

While discussing qualitative content analysis such as semiotics and discourse analysis it was argued that the search for meaning can be both simple and complex because meaning itself can be apparent or encoded in symbolism and associations.

We also detailed the procedure to be followed while using this method - right from the statement of the research problem to defining the population, selecting a sample, constructing a categorisation system, selecting a unit of analysis and coding the content to arrive at certain conclusions and inferences.

8.7 REFERENCES AND FURTHER READINGS

Berelson, B. (1952), *Content Analysis in Communication Research*, New York Free Press.

Berger, A.A. (1990), *Media Analysis Techniques*, Newbury Park, CA: Sage.

Kracauer, S. (1952), The Challenge of Qualitative Content Analysis. *Public Opinion Quarterly*, 16:631-42.

Krippendorff, K., (1980), *Content Analysis: An Introduction to its Methodology*, Newbury Park, London: Sage.

Neuman, W.L., (2007), *Social Research Methods - Qualitative and Quantitative Approaches*, Pearson Education. N.D.

Propp, V., (1975 [1928]), *Morphology of the Folk Tale*, London: Austin.

Radway, J., (1984), *Reading the Romance*. Chapel Hill: University of North Carolina Press.

Wimmer, R.D and Dominick, J.R, (2003), *Mass Media Research - An Introduction*, Thomson Wordsworth Publishers, London.

8.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Content analysis is a research technique for the objective, systematic and quantitative description of the manifest content of communication.
- 2) Bernard Berelson, author of book is 'Content Analysis in Communication Research' is known as the father of Quantitative Content Analysis.
- 3) Empirical inquiries into communications content date back to the 1960s when the Church began to be increasingly concerned with 'non-religious' content as carried by the newspapers of the day.

Check Your Progress 2

- 1) Content Analysis is used to describe communication content and to establish a starting point for media effects studies.
- 2) The content to conduct certain studies might not be available either because it was destroyed or because it is classified. It can also be a painstaking and time consuming process.

Check Your Progress 3

- 1) The basic difference between the two is with regards to the location of meaning. In the former case the researcher searches for 'fixed' meanings located on the surface. In the latter case the researcher searches for 'multiple' meanings located beneath the surface.
- 2) In Semiotic Analysis meaning is determined by dividing content into two components - the signifier and the signified - and by studying the relationship between the two. Meaning is also understood in terms of what is not signified. For instance, by showing candlelight, roses, a dinner table set for two, the film director is signifying a romantic evening. Simultaneously s/he is signifying that it is not a relationship of a general nature. Meaning is also understood in terms of the associations that a signifier word or image has - in certain cultures white signifies widowhood while in other cultures it is the colour of a bride's wedding gown.
- 3) Rhetorical Analysis

Check Your Progress 4

- 1) The categories should be exhaustive, mutually exclusive and reliable.
- 2) The four levels of measurement are: nominal, ordinal, interval and ratio.
- 3) Multi-stage sampling.

UNIT 9 EXPERIMENTAL METHOD

Structure

- 9.0 Introduction
- 9.1 Learning Outcomes
- 9.2 Nature of Experimental Method
- 9.3 Classic Experimental Research Design
- 9.4 Process of Experimental Research
 - 9.4.1 Control of Extraneous Variables
 - 9.4.2 Randomisation
- 9.5 Experimental Design
 - 9.5.1 Factorial studies
 - 9.5.2 Quasi experimental design
- 9.6 Field Experiments
- 9.7 Merits and Demerits of Experimental Method
- 9.8 Let Us Sum Up
- 9.9 Keywords
- 9.10 References and Further Readings
- 9.11 Check Your Progress: Possible Answers

9.0 INTRODUCTION

Communication research has established itself on a firm footing and is a fertile field for young researchers. The area is rich, vast and varied. Experimental method is a well-established method of media research but its use is restricted. There is a tendency among researchers to opt for the survey method. Experimental research is highly popular in psychology and social psychology. In the case of media research, experimental method can produce an enormous amount of data. However, media researchers find several issues for conducting experiments. For instance, the method is largely quantitative and uses extensive statistical procedures for analysis. Issues relating to infrastructure, i.e. getting the appropriate testing conditions, and/or getting matching subjects for experiments pose problems for researchers. There can be laboratory experiments or field experiments. The purpose is to find out and establish a causal relationship between an independent variable and a dependent variable. This method is also amenable to statistical analysis depending on the data generated.

We have discussed earlier that the selection of a research method depends upon the topic selected and the researcher's methodological approach towards the selected topic. The experimental technique supports empirical approach to research.

9.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- discuss the nature and method of experimental research;

- describe the process of classic experimental research design;
- differentiate between experimental and quasi-experimental designs;
- explain the nature and scope of field experiments; and
- outline the advantages and disadvantages of experimental method of research.

9.2 NATURE OF EXPERIMENTAL METHOD

As already stated, the experimental method is widely used in natural sciences and in certain social science disciplines. Being quantitative in nature, experiments attempt to find out the relationship of cause and effect between factors. In other words, they help establish and find out cause and effect. Of course, there are differing views. Some scholars think it is not easy to find out a cause and effect relationship between two variables, independent and dependent. Nonetheless, the experimental method is one of the effective methods in which the researcher has the opportunity to control the time order when presenting two variables. This enables him/her to ensure that cause happens first and then the effect and s/he can also control other possible causes of the variable being studied.

Experimental method is scientific in nature and character. As science makes progress, its forms of explanations also keep changing. Since social sciences adopt scientific methods of empiricism, their explanations of social phenomena also keep changing. Explanations are categorised into deductive and probabilistic. This classification is done on the basis of types of generalisations that an explanation employs. Let us take the deductive explanation for evaluation. The pre-conditions of a deductive explanation are;

- 1) A universal generalisation.
- 2) A statement of conditions which makes the generalisation appears true.
- 3) An event to be explained (explicandum), and
- 4) The rules of logic.

The law of gravitation is a good example for a deductive explanation. Its applicability encompasses past, present and future. In natural sciences, deductive explanations are made possible with manipulation and control through experimentation. Experiments help us demonstrate the effect of one phenomenon on another phenomenon. However, only few phenomena of interest to social scientists are susceptible to manipulation and control. Yet there are certain ethical questions related to the manipulation of social phenomena by social scientists. Besides, most human factors are not amenable to the research methods adopted by natural scientists.

Social scientists mostly prefer to have probabilistic explanations. In these, a generalisation is supposed to provide evidence for the phenomena under investigation. No explanation can be deduced from the generalisation. In universal generalisation, we may say 'All A is B', whereas a probabilistic generalisation will say that 'n percent of A and B'. Probabilistic explanations are weak compared to deductive explanations. Probabilistic explanations have a very limited predictive capacity. The deductive explanation is strong in its predictive power.

9.3 CLASSIC EXPERIMENTAL RESEARCH DESIGN

A classic experimental research design contains two groups for comparison. One is an experimental group and the other is a control group. These two groups are equal in every respect. However, there is one difference. The experimental group is administered or exposed to an independent variable, which is called experimental stimulus, and the control group is not exposed to the experimental stimulus. Measurements on the dependent variable are called scores and are taken twice from each group. The first measurement is known as the **pre-test**. It is taken prior to the administration of experimental stimulus to the experimental group. The control group also undergoes measurement. The second set of measurement takes place after the experimental group is exposed to the experimental stimulus. This is known as the **post-test**. The control group, of course, is not exposed to the experimental stimulus. But both the groups are measured. The difference in measurement between the post-test and pre-test will be compared in each of the two groups, experimental and control. In case of the difference in the experimental group, we can safely assume that the independent variable causes changes in the dependent variable, or the independent variable has an effect on the dependent variable and is causally related dependent variable.

Group	Pre-test	Post-test	Difference
Experimental	M1	M2	$M2 - M1 = R_a$
Control	M3	M4	$M4 - M3 = R_b$

Selden Menefee and Audrey Menefee who visited Tiptur area of Karnataka conducted an experimental study using a mimeographed newspaper to measure the information level of sample subjects chosen. When the measurements were compared between the experimental group and the control group, it was found that the information level of the experimental group had gone up due to the exposure to the mimeographed newspaper more than the control group that was not exposed to the newspaper, the experimental stimulus. This meant that exposure to the newspaper caused the increase in the information level of those exposed to it.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What is the main purpose of experimental method?

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2) Identify the pre-conditions of a deductive explanation.

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- 3) Briefly explain the classical experimental method.

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9.4 PROCESS OF EXPERIMENTAL RESEARCH

The experimental method involves two operations: manipulation and observation. In scientific jargon, researchers manipulate an independent variable and then observe the consequence or response of the subjects as to the dependent variable. Each experiment may be different, but every researcher has to consider the common steps to follow in an experiment.

Below is a direct quotation from Wimmer and Dominick (2003) who have identified the steps as:

- 1) “Every researcher has to select the setting. Most experiments are conducted either in a laboratory or any other environment which will be under the control of the investigator. Experiments are also conducted under natural settings, but the researcher will have limited control over his experiment.
- 2) Any research will have to select the experimental design. The hypotheses or research question will determine the appropriateness of the design. The types of variable available for manipulation or evaluation, availability of respondents, and resources will impact the design selection.
- 3) Operationalise the variables: In experimental research, the independent variables are operationalised after their creation through manipulation. Scales or rules are constructed to categorise observations of behaviour to operationalise the dependent variables.
- 4) Decide how to manipulate the independent variable. A set of clear and specific instructions, events or stimuli is developed to manipulate the independent variables and present them to the subjects of the experiment. The two types of manipulations are straight forward and staged. “In straight forward manipulation, written materials, verbal instructions, or other stimuli are presented to the subjects”. If it is staged manipulation, the investigators create events and circumstances that help them to manipulate the independent variable. It can be simple or elaborate. The use of a confederate is also in vogue. A confederate is a person who acts like a subject, but in reality, he is also part of the manipulation.
- 5) Select and assign subjects to experimental conditions. In order to guarantee external validity, the best way is to select the subjects randomly from the population and assign them to the experimental and control groups.
- 6) Conduct a pilot study: A pilot study involving a small number of subjects will reveal problems and enable the investigator make suitable corrections and revisions. It is done to find out whether the manipulation of the independent variable has had the desired effect.

- 7) Administer the experiment: The main phase of research is data collection. Experimental manipulations are done on the subjects, either individuals or groups. Then the effect of independent variable is measured and the debriefing of subjects takes place. While debriefing the investigator will explain the objectives and consequences or implications of the study undertaken.
- 8) Analysis and interpretation of results: The scores of subjects on the dependent variables measured are tabulated and statistically analysed. Then the investigator will interpret the results.”

9.4.1 Control of Extraneous Variables

In media research, an investigator comes across unwanted extraneous variables. S/he has to remove or control them to ensure internal validity of the instrument/ method employed for research. These extraneous variables are also referred to as **confounding variables**, which normally contaminate and complicate the findings. The confounding variables are controlled by the environment, experimental manipulations, experimental design and assigning subjects to both experimental and control groups.

Generally, three types of techniques are used in assigning subjects to groups. They are:

- Randomising
- Matching
- Generalisation

Matching and randomisation are the two methods of control. Matching controls variables are known to the researcher before the research process starts. Randomisation will enable him/her to remove the effect of unexpected or not-seen-before factors.

Matching is a method to equate the experimental group with control group, on external variables presumably related to research hypotheses. The two methods of matching the groups are precision matching and frequency distribution. Precision matching is also known as pairwise matching. Each subject in the experimental group is matched by another one in the control group with the same characteristics. If, for example, one person is of 30 years of age in the experimental group, another subject in the control group will also be of the same age. The problem with this method is the matching of a large number of variables, which is difficult. Pairwise matching is a tough task. Precision matching has a very high number of losses of cases in the absence of compatible twins.

This is usually substituted by frequency distribution. Here experimental and control groups are matched for each variable separately rather than in combination. It is not one-to-one matching, but these two groups are matched on identified central characteristics. For instance, the average age of one group should match with the average age of another group. This may be less precise, but is easy to manage. As we know, matching can control a limited number of pre-identified extrinsic variables. There may be a good number of confounding variables that affect the relationship between independent and dependent variables.

9.4.2 Randomisation

One cannot be sure of eliminating the confounding variables. Hence, there is scope for wrong interpretation of results. This can be avoided by opting for randomisation - a process by which subjects are assigned to the experimental and control groups. The assignment can be done by flipping the coin, or by using a table of random digits or the lottery method or any other suitable method. This will ensure an equal probability of being selected to either the experimental or the control group. Randomisation will remove any systematic mistake because of extrinsic variable that may be related to either independent or dependent variables.

Internal validity is important in communication research. Another equally significant aspect of research is the generalisability of the findings. The effect of one variable on another in a natural setting and on a bigger population is known as external validity of research designs. It also refers to the ability to generalise the research findings. Two factors related to external validity that can affect the generalisation of research results are representativeness of the sample and the reactive arrangements.

The random assignment of subjects to the experimental and control groups will ensure equality between the groups and also internal validity of the investigation. However, it may not be representative of the larger population, but represent only the sample selected. High rate of refusal and enormous cost may affect opting for a highly representative sample. Next will be the reactive arrangements. The results of a study should not only be generalised to a large population, but also to a real-life situation. This is difficult to accomplish. It may result in reactive effect of testing on the subjects. For instance, a pre-test may affect the response of subjects to the experimental stimulus. Such an effect can be avoided by opting for post-test only.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Explain the two operations involved in experimental method.

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2) List the common steps to follow in experimental research.

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- 3) Explain in brief the three factors that can be used to control the confounding variables.

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9.5 EXPERIMENTAL DESIGN

The experimental design refers to the ‘total experimental plan or structure of the research’. It is selecting and planning the experiment related to a research problem. The simple procedure is pre-test, experiment and post-test. Other factors like variables, sampling, control and constructing a measuring instrument should also be included in the process. Normally, research designs are classified as experimental and quasi-experimental. We have already discussed the basic procedure of a classical research design. Some of the popular experimental designs are recounted here:

a) Pre-test-post-test control group

The pretest-posttest control group design is one of the basic procedures commonly adopted by media researchers. This procedure helps in controlling the rival hypotheses because of effects of maturation, testing, history and other sources, because each group faces the same situations during the experiment. This is the other name for classical experimental design.

b) Post-test only control group

There is a possibility of subjects becoming sensitive to the experiment during pre-test. In the post-test only control group method, neither the experimental group nor the control group is exposed to the pre-test. Only the experimental group is exposed to the independent variable and these will be a post-test of the group. Then the two groups are compared to the presence of statistical significance. This helps in controlling the rival explanation as both groups are equally affected by factors like maturation, history and others.

c) Solomon four-group design

We have already observed that a pre-test has its own merits and demerits. It enables us to assess the time sequence and also it is the basis of comparison. Yet we may have reactive effects in the process. More than that, these are occasions where we may not be able to have the luxury of pre-measurement period. The Solomon four-group design will have the same features of a classic experimental research design. The difference is that it will have an additional set of control and experimental groups that are not made subject to pre-testing. When we have two experimental and control groups each, the comparisons will show whether the experimental stimulus (x) has independently affected the groups that were not made sensitive by a pre-test. If we find that the experimental stimulus had its effect without pre-test, the findings can be generalised to the target population. The element of causality will increase in its strength.

9.5.1 Factorial studies

When a researcher analyses two or more independent variables simultaneously, it is called factorial design. In this, each independent variable is called a factor. The factorial study has advantages like saving of time, money and resources. It allows the investigator to know the interaction between two or more inter-dependent variables in the effects they cause on the dependent variable. If we say that it is a two-factor design, it means that there are two independent variables which are manipulated. The same is the case with a three-factor design where three independent variables are manipulated. Factors can have two or more levels. An example here will make its understanding easy. Imagine that there is an advertising campaign on radio and television. The subjects of these media are assigned two groups, one that is exposed to the radio and the other, exposed to television. These are known as two levels of investigation.

9.5.2 Quasi Experimental Design

The quasi experimental design is not suitable for application in certain research situations. This is especially true when manipulation and random assignment of respondents are not possible. In social sciences, two types of relationships are identified. The first one is the stimulus-response, the second one is property-disposition relationship. For the first one, reward and satisfaction or advertisement and consumption levels can be given as examples. For property-disposition relations attitudes, values and orientations can be given as examples. Stimulus-response situations are best suited for experimental methods while property-disposition can be tackled by quasi-experimental designs.

The four areas of difference between these two types of designs are identified.

- 1) The time gap between the effect of independent variable and the response to it. In stimulus - response relations, the time interval is short. In contrast, in property disposition relations it is longer.
- 2) A stimulus is easy to isolate and its effect can be well-described . On the other, properties like social class are unclear and dispositions like prestige and education have their own effect on other elements.
- 3) The, 'before and after' comparison is easy in experimental studies while it is difficult in quasi-experimental studies. For example, two groups with different properties and dispositions cannot be compared. An upper-class group differs from a lower-class group on many counts, like values, and social practices.
- 4) The time sequence of event varies in experimental research, the direction of causality is clear for pre and post comparisons. However, the time sequence is difficult to establish in the case of property-disposition relationship. As such, manipulation, comparison and control cannot be applied to property disposition relationships easily.

Experimental methods are strong on control but weak on representation. Quasi-experimental studies, particularly surveys, are strong on representation, but weak in control. Experiments can control both extrinsic and intrinsic variable through randomisation. They also have control over the introduction of experimental stimulus. This helps them in determining the direction of causation. These

advantages cannot be seen in quasi-experimental designs. On the other, laboratory experiments cannot replicate social life. Quasi experiments will not have the random assignment of subjects to experimental groups. The major types of quasi-experimental designs are: pretest-posttest non-equivalent control group design and interrupted time-series design. The one-shot case study, correlational designs, panel and trend studies are also considered as types of quasi-experimental studies by some scholars though they can be longitudinal studies.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) Explain the Solomon four-group methods.

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2) Distinguish between stimulus-response and property-disposition relationships.

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3) List the four areas of difference between experimental and quasi-experimental designs.

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9.6 FIELD EXPERIMENTS

As you are aware, the laboratory experiments are artificial, as such, they do not have external validity. In contrast, field experiments have external validity. Subjects react in their natural setting with normal behaviour. For example, watching a TV commercial in a laboratory setting is different from viewing it in a natural setting. One of the characteristics or advantages of field experiments is that it is closer to reality and often non-reactive. Complex social processes and events can be studied. Field experiments can be less expensive, as they do not need special equipment. Yet, if the size of the study increase then the expenditure may also increase. Sometimes, field experiment is the only option left in certain social situations.

There are a few practical difficulties with the field experiments. For instance, certain research studies cannot be done in the field due to ethical reasons. The impact of television violence on children cannot be measured through field experiments. Field experiments now and then face external obstacles that may be unexpected. It is time-consuming when the researcher wants to establish contacts, seek co-operation and obtain necessary permission. Often, this procedure takes weeks and months. The most difficult part of field experiment is the control of intervening variables. The extraneous variables will affect the accuracy of experiments, and in turn the entire research work.

Types of field experiments

Field experiments have been categorised into two: experiments in which the investigator manipulates the independent variable and the other in which manipulation of independent variable happens naturally. Wimmer and Dominick provide an example for the first type. Imagine an investigator wants to know the effects of not being able to read a newspaper. The option is to choose two samples comparable and not allow one sample to read a newspaper for a period of time. The second sample will continue to read the newspaper. Later a comparison will be made to know the effect on the other areas of life like interpersonal communication. The other types of field experiment deal with passive manipulation of the experimental stimulus. In some cases of field experiment, the investigators may not be able to assign subjects randomly to experimental and control groups. Such field experiments are known as quasi experiments. One major challenge for a researcher is to guard against alternative explanations.

9.7 MERITS AND DEMERITS OF EXPERIMENTAL METHOD

Like every method of research in social sciences, the experimental method has its own advantages and disadvantages. The utilitarian advantages of experiments are:

- a) **Evidence of causality:** Experiments enable us to find out the cause and effect relationship between an independent variable and a dependent variable. They also support in controlling the time order besides other possible causes that research effort.
- b) **Control:** As already stated, control is an advantage of the experiments. An investigator will control the environment, and isolate the testing from other influences and effects. It means they can structure the experimental setting according to their choice. However, the artificial nature of the laboratory setting is a disadvantage. Selection and control of subjects is another advantage in experimental research.
- c) **Cost:** When compared to other research methods, experiments can be less expensive.
- d) **Replication:** The experiments can be replicated. Since the conditions under which experiments are conducted are clearly described, it becomes easy to repeat the experiments by others. Modifications can also be done.

Disadvantages:

The experimental method has its own disadvantages:

- a) **Artificiality:** As already stated, artificiality of environment of research as a major problem in experiments as behavioural alterations take place when media researchers employ the laboratory setting. Because of artificial environment in the laboratory, results that emanate from these do not have any resemblance when compared to the behaviour in natural atmosphere or real life situation. In real life, subjects are exposed competing stimuli from other sources. The artificiality may affect the external validity of research.
- b) **Researcher bias:** The experiment can influence the experiment so that s/he can get the result according to the pre-decided hypotheses. In order to avoid this type of bias on the part of researcher, a technique called double-blind is used. In this, neither the researchers nor the subjects will know whether a particular subject is given to the experimental or control group.
- c) **Limited scope:** It is a fact that only select research questions can be experimented upon. When issues and topics in mass communication concern a huge number of people, experiments on a massive scale are not possible to conduct.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What are the characteristics of field experiments?

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2) Identify the categories of field experiments.

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3) Enumerate the advantages and disadvantages of experimental technique.

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9.8 LET US SUM UP

Experimental method is a technique used in communication research. However, its adoption is restricted in media research unlike in psychology and social psychology. Natural sciences make the best use of experimental research. Experiments attempt to find out the relationship between factors of cause and effect. These enable us to ensure that cause happens in a scientific nature. Explanations in social sciences are classified as deductive and probabilistic.

Deductive explanations are strong as their applicability encompasses past, present and future. In natural sciences, deductive explanations are made possible with manipulation and control. But in social sciences only a few phenomena are amenable to manipulation and control.

A classic research design contains two equal groups: experimental and control. The experimental group is exposed to experimental stimulus and the control group is not. The measurements taken during pre-test and post-test are compared to know whether the experiment involves manipulation and control, which are difficult to achieve in social sciences. The experimental technique will normally have a process of eight steps. Control of confounding or extraneous variables can be done by matching, randomisation and generalisation. Internal and external validity will have to be achieved in experimental research. There are two designs when we deal with experimental method: Experimental design and quasi-experimental designs. Pretest-posttest control group, post-test only group and Solomon four-group designs belong to experimental design.

When a researcher analyses two or more independent variables simultaneously, it is called a factorial study or design. Here each independent variable is called a factor. Pretest-posttest non-equivalent control group design and interrupted time series design belong to the category of quasi experiments.

Experimental designs are not suitable for application in certain research situations, especially when manipulation and random assignment of respondents are not possible. Experimental designs are best suited to stimulus-response relationships but not for property-disposition relationships. The property-disposition relationships can be tackled by quasi-experimental designs that include one-shot case study, correlational designs, panel and trend studies. Field experiments are conducted in natural settings. They are closer to reality and often non-reactive. They do not need any special equipment. The costs can be less compared to other methods. The advantages of experimental methods are evidence of causality, control, cost benefit and replication. The disadvantages are artificiality, researcher's bias and limited scope.

9.9 KEYWORDS

Confounding variables: Variables which are extraneous to the experiment. They contaminate the findings.

Deductive explanation: An explanation that has the element of generalisation regarding past, present and future.

Empiricism: Verification with strong evidence.

Explicandum: An event or phenomenon to be explained.

Manipulation: Deliberately produce a change in the phenomenon selected.

Pilot study: Involves a small number of subjects on whom the study is conducted.

Subjects: People who are selected to participate in the experiment.

Verstehen: A tradition of understanding human behaviour.

9.10 REFERENCES AND FURTHER READINGS

Broota, K.D (1992), *Experimental Design in Behavioural Research*, New Delhi, Wiley Eastern Limited.

Nachmias, David and Chava Nachmais (1976), *Research Methods in the Social Sciences*, New York, St. Martin's Press.

Priest, Susanna Hornig (2010), *Doing Media Research: An Introduction*, New Delhi:, SAGE Publications India Pvt. Ltd.

Wimmer D. Roger and Joseph R. Dominick (2003), *Mass Media Research*, Belmont (USA) Wadsworth/Thomson Learning.

9.11 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) The main purpose of the experimental method is to find out the relationship between factors of cause and effect. Experiments are more suitable to natural sciences than social sciences. In other words, experimental research attempts to find out the cause and effect relationship between an independent variable and a dependent variable.
- 2) The four pre-conditions of a deductive explanation are:
 - A universal generalisation.
 - A statement of conditions which makes the generalisation appear true.
 - An event to be explained (explicandum)
 - The rules of logic.
- 3) A classical experimental research design has two equal groups for comparison: experimental and control. The experimental group is exposed to the experimental stimulus, but the control group is not, for post-test. The difference between post-test and pre-test on the measurements taken will prove that the independent variable has an effect on the dependent variable.

Check Your Progress 2

- 1) The two operations involved in experimental research are manipulation and observation. In scientific jargon, researchers manipulate an independent variable and then observe the response of the subjects on the dependent variable.
- 2) Selection of the setting, selection of the experimental design, operationalisation of variables, manipulation of independent variable, selection and assignment of subjects to experimental conditions, conducting

a pilot study, administering the experiment and analysis and interpretation of extraneous variables are the eight steps involved in experimental research.

- 3) The confounding variables can be controlled by: matching, randomisation and generalisation, Matching can be either precision matching or frequency distribution. Randomisation can be done by assigning the subjects to experimental and control groups by using different methods. If the findings can be applied to a large population, it is called generalisation.

Check Your Progress 3

- 1) The Solomon four-group design will have the same features of classic experimental research design, except that it will have an additional set of control and experimental set of control and experimental groups that are not subjected to pre-testing. If we find that the experimental stimulus had its effect without pre-test, the findings can be generalised.
- 2) The stimulus-response relationship is based on reward-satisfaction proposition. The property-disposition relationship has its basis in attitudes, values and orientations. The first one amenable to experiments, while the property-disposition relations are taken case of quasi-experimental designs.
- 3) The four areas of difference between experimental and quasi-experimental designs are: time gap between the effect of independent variable and the response to it; a stimulus can be easily isolated but properties are not; before and after comparison; and the variation in time sequence.

Check Your Progress 4

- 1) Field experiments are being carried out in natural settings. Because of this, they have external validity. Subjects will react with their normal behaviour. Field experiments are non-reactive and less expensive and do not need special equipment or facilities.
- 2) There are two types of field experiments: Experiments in which the researcher manipulates the independent variable and the other in which manipulation of the independent variable happens naturally.
- 3) The advantages of experimental method are: evidence of causality, control, less cost, and replication. The demerits are artificiality, bias of the researcher and limited scope.